

FAKE NEWS DETECTION USING NLP, TF-IDF AND BERT

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Section - 4

Department- CSE

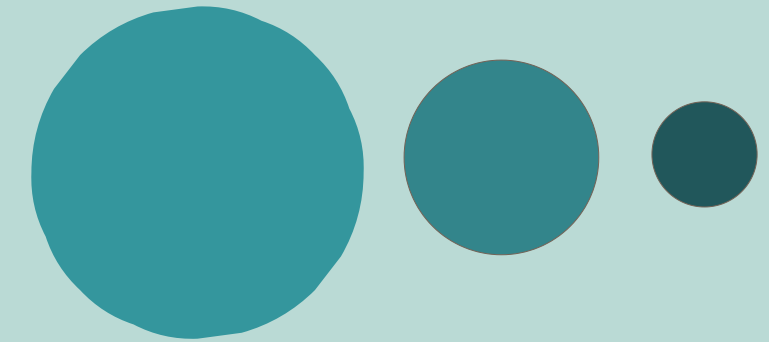
**Technologies Used: Python, NLP, TF-IDF, Logistic Regression,
BERT, Gradio**



Introduction

- Fake news is false or misleading information presented in the form of news. It can spread rapidly through digital platforms and make it difficult for users to distinguish between genuine and false information.
- Our project develops a machine learning-based Fake News Detection system that analyzes the textual content of news articles and classifies them as Fake or Real.
- The system uses:
 - NLP for text preprocessing
 - TF-IDF for converting text into numerical features
 - Logistic Regression for classification
 - BERT for advanced contextual analysis
 - Gradio for the user interface

Problem Statement



Problem

- Fake news spreads rapidly through digital platforms.
- Fake and real news can have similar writing styles, making manual identification difficult.
- Traditional text-based models such as TF-IDF-based classifiers may not capture deeper contextual relationships.
- Advanced models such as BERT provide contextual understanding but require greater computational resources.
- Therefore, there is a need for a practical system that compares lightweight and contextual approaches and provides an easy-to-use fake/real prediction interface.

Objective

To develop an end-to-end NLP-based fake news detection system using TF-IDF, Logistic Regression, SVM and BERT, evaluate their performance, and provide predictions through an interactive frontend.

Literature Survey

Literature Survey

Author(s) & Year	Technique / Approach	Dataset	Key Findings	Limitations
Shu et al. (2017)	Feature-based ML (SVM, Logistic Regression)	FakeNewsNet	Used linguistic and social features for fake news detection.	Limited contextual understanding.
Devlin et al. (2018)	BERT (Transformer Model)	Various news datasets	Showed that pre-trained language models capture deep contextual meaning and improve accuracy.	High computational cost and training time.
Rashkin et al. (2017)	Neural Networks (LSTM)	LIAR dataset	Used deep learning for text classification and achieved good performance.	Requires large data and high resources.
Granik & Mesyura (2017)	Traditional ML (Naive Bayes, SVM, RF)	Collected news dataset	Compared multiple classifiers and showed reasonable results with simple features.	Struggles with subtle and context-dependent fake news.
Vosoughi et al. (2018)	Propagation-based features	Twitter dataset	Analyzed how fake news spreads faster than real news.	Focused mainly on social media data.
Kaliyar et al. (2021)	Hybrid approaches (TF-IDF + Deep Learning)	Multiple datasets	Combination of traditional and deep learning methods improves detection performance.	Not a complete end-to-end system.

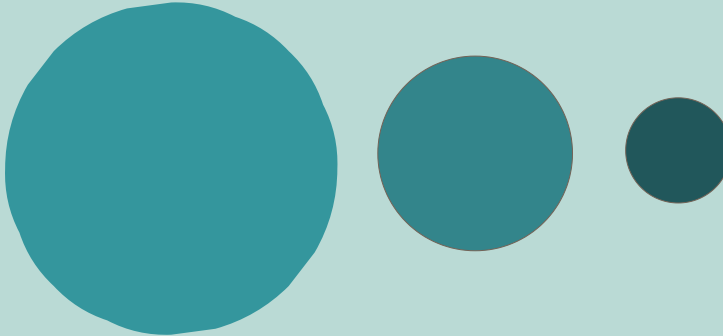
Research Gap

From the literature, it is observed that most existing studies focus on either traditional machine learning methods or advanced deep learning models for fake news detection. While traditional models are simple and efficient, they lack deep contextual understanding. On the other hand, advanced models like BERT provide better contextual analysis but require high computational resources. In addition, many studies do not provide a complete end-to-end system with an interactive interface.



Identified Research Gaps

-  Most studies use only one or a few algorithms without comparing lightweight and advanced models.
-  Traditional text-based models do not capture deeper contextual meaning.
-  Advanced models like BERT have higher computational requirements.
-  Many existing works do not provide a user-friendly system for real-time fake news prediction.

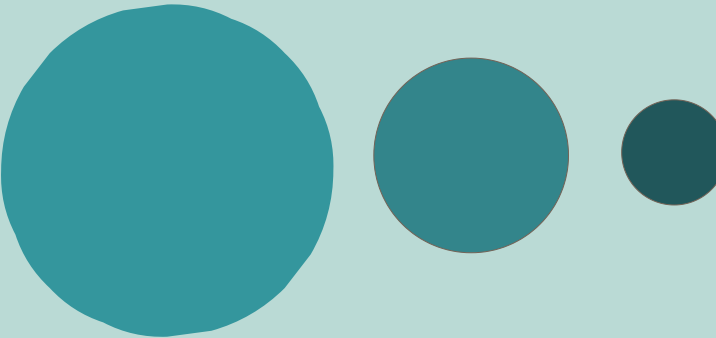


Dataset Description

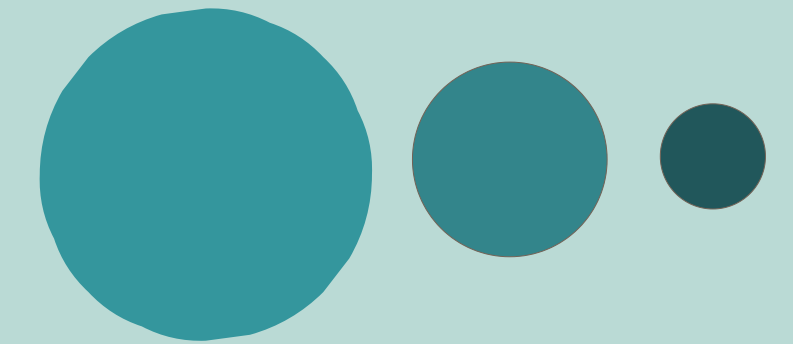
Dataset: Fake News Detection Dataset	
Property	Details
Total records	2000
Total columns	4
Classes	2
Fake News	1000
Real News	1000
Missing values	0
Duplicate records	0

Dataset Columns

- Columns
- serial_number
- title
- text
- label
- Label Mapping:
- 0 → Fake News
- 1 → Real News
- The dataset specifies 2,000 records with balanced fake/real classes and no missing or duplicate records.



Exploratory Data Analysis (EDA)



We performed EDA to understand the dataset before training the models.

The analysis includes:

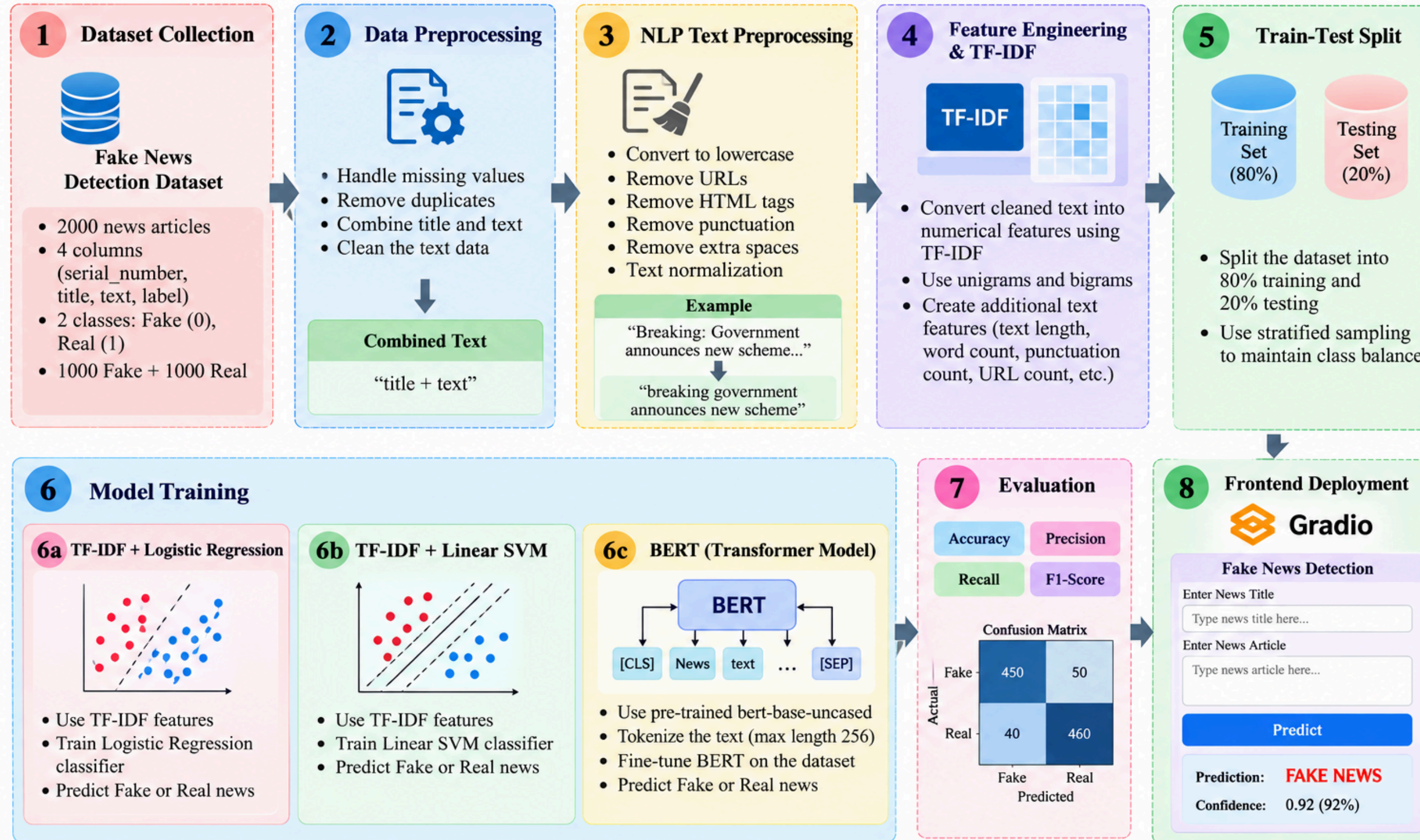
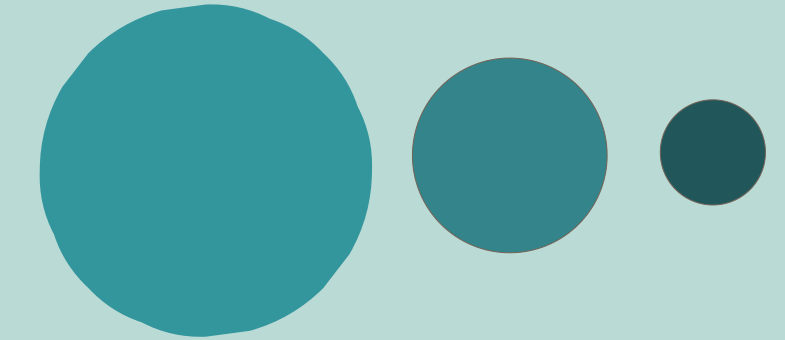
- Class distribution
- Text length analysis
- Word count analysis
- Fake vs Real text comparison
- Feature correlation
- Feature distributions
- Sensationalism analysis

Important Observation

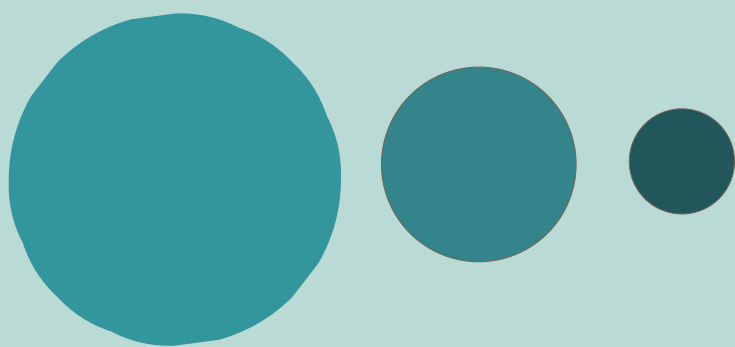
The dataset contains an equal number of Fake and Real news samples, providing a balanced classification problem.

Methodology

Methodology



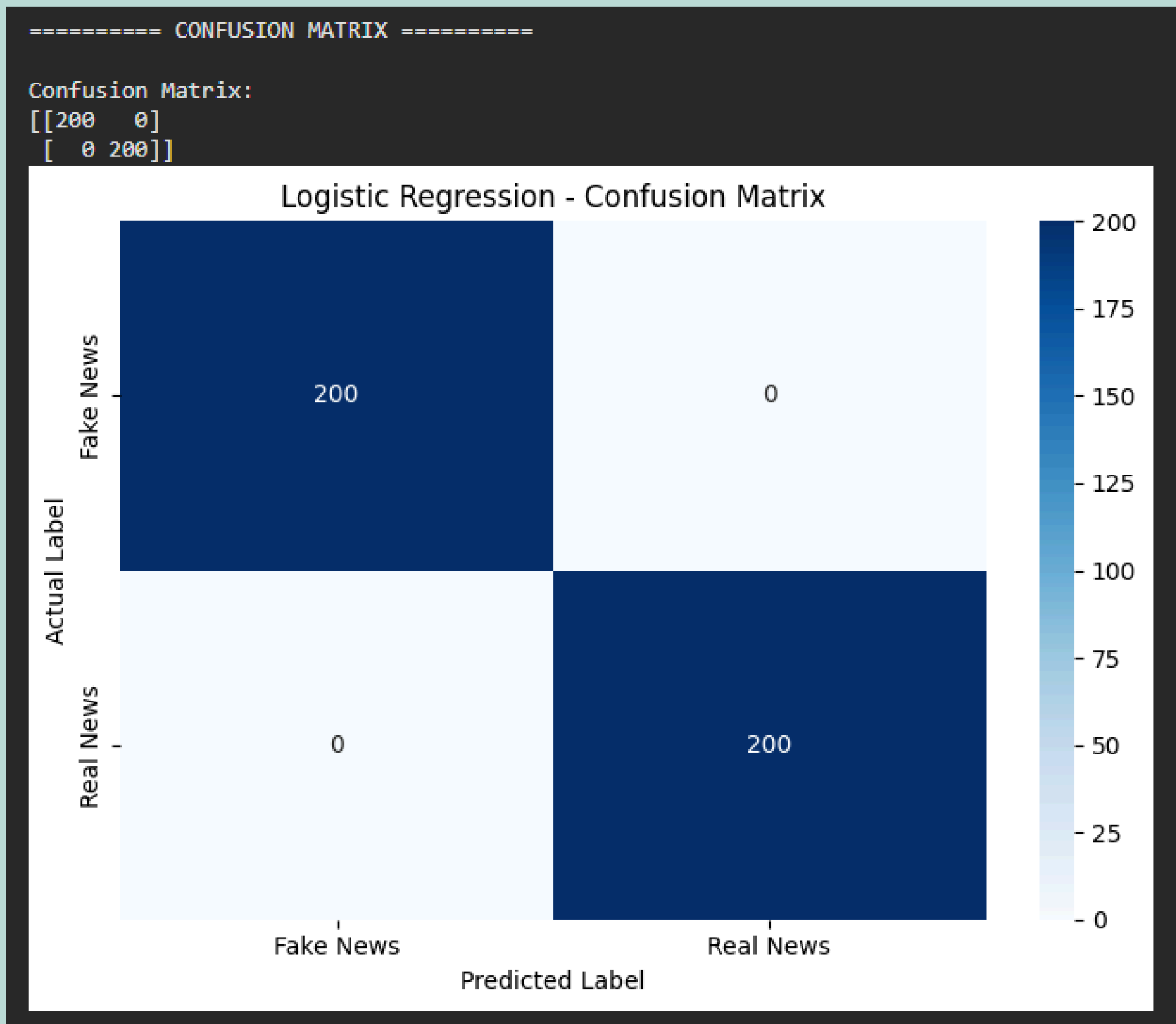
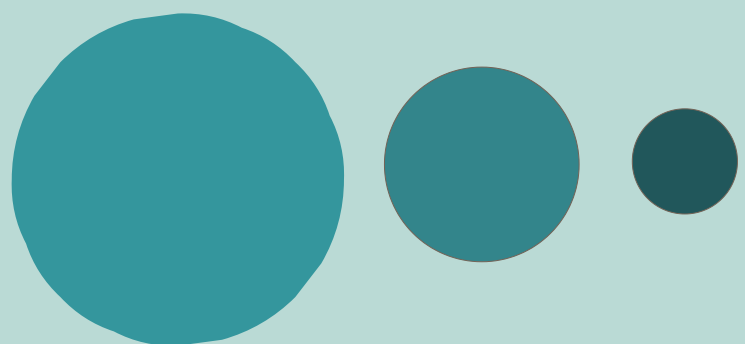
Results and Discussion



Model	Accuracy	Precision	Recall	F1-Score 
TF-IDF + Logistic Regression	100.0%	100.0%	100.0%	100.0%
TF-IDF + Linear SVM	100.0%	100.0%	100.0%	100.0%
BERT	100.0%	100.0%	100.0%	100.0%

Use these four metrics:	
Metric	Meaning
Accuracy	Overall correct predictions
Precision	Correct positive predictions among predicted positives
Recall	Correct positive predictions among actual positives
F1-Score	Balance between precision and recall

Confusion Matrix



- **True Positive (TP):** Fake news correctly identified as Fake.
- **True Negative (TN):** Real news correctly identified as Real.
- **False Positive (FP):** Real news incorrectly classified as Fake.
- **False Negative (FN):** Fake news incorrectly classified as Real.

Output Demonstration

News Title

NASA discovers a new planet with conditions similar to Earth

News Article

Scientists have announced the discovery of a distant planet that may have conditions suitable for liquid water. Researchers are continuing to study the planet using advanced telescopes to understand its atmosphere and composition.

CHECK NEWS

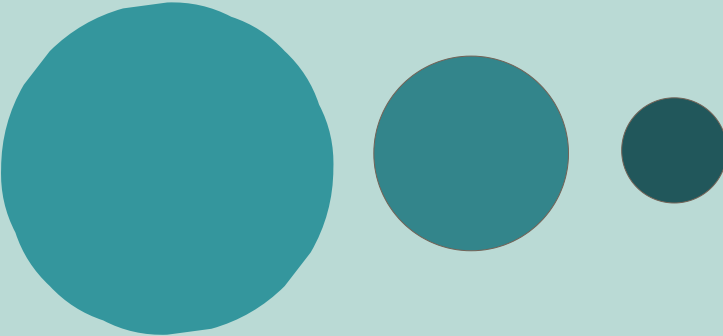
Prediction Result

Prediction

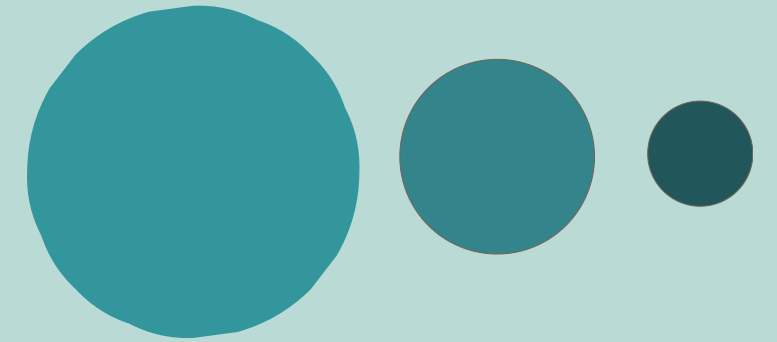
☒ REAL NEWS

Confidence

87.84%



Conclusion & Future Scope



Conclusion

- Developed an NLP-based fake news classification system.
- News text is cleaned and preprocessed using NLP techniques.
- TF-IDF converts text into numerical features.
- Logistic Regression, Linear SVM and BERT are used for classification and comparison.
- Models are evaluated using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.
- A Gradio frontend provides Fake/Real predictions with confidence.
- The system is designed as a supportive tool for news analysis.

Future Scope

- Larger and more diverse datasets
- Real-time fake news detection
- Web/mobile deployment
- Multimodal detection using text + images
- Social-media information analysis
- Improved BERT/Transformer fine-tuning
- Explainable AI for prediction reasoning



Thank You

